

R1 = first try , R2 = re-run

Q2 — Feature Statistics

Metho d	Shape	Top variance feature	2nd	3rd	Top corr pair	lrl
Baseli ne RF	11020 ×30	global_min_ mean_z (0.357)	rolling_mean _mean_x (0.329)	global_mean_ mean_x (0.328)	global_mean _mean_x / rolling_mean _mean_x	0.9 999
Hybrid RF + CNN- LSTM	11020 ×30	<i>(stale globals)</i>			global_mean _mean_x / rolling_mean _mean_x	0.9 999
FFT + CNN- LSTM	11020 ×30	<i>(stale globals)</i>			global_mean _mean_x / rolling_mean _mean_x	0.9 999
SMA + peak- freque ncy + CNN- LSTM	11020 ×30	<i>(stale globals)</i>			global_mean _mean_x / rolling_mean _mean_x	0.9 999
BEST RF + LGB + Residu al CNN	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895
LGB + Residu al CNN + augme	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895

nted CNN ensem ble						
LightG BM 5- fold + single CNN	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895
LGB + XGB + multi- seed CNN extend ed feature s	11020 ×133	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	fft_x_peak_a mp / fft_x_top1_a mp	0.9 902
LGB + XGB + multi- seed CNN extend ed feature s	11020 ×133	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	fft_x_peak_a mp / fft_x_top1_a mp	0.9 902
RF + LGB 5-fold + multi- seed CNN	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895

RF + LGB + Residual CNN weight boosted	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895
CNN seed search	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895
Person - independent LightGBM normalized features	11020 ×112	n_x_fft_spect ral_energy (3841.4)	n_y_fft_spect ral_energy (3639.2)	n_z_fft_spectr al_energy (3280.1)	n_z_kurt / m_z_kurt	1.0 000
Bigger CNN + label smoothing + spatial dropout	11020 ×69	fft_y_spectral _energy (1.94e7)	fft_z_spectral _energy (1.86e7)	fft_x_spectral _energy (1.45e7)	mag_mean / mag_energy	0.9 895

Q2 — Feature Importances (top 5, RF/tree models only)

Method	#1	#2	#3	#4	#5
Baseline RF	global_mean _std_z (0.108)	global_mean _std_x (0.096)	rolling_mean _std_x (0.082)	rolling_mean_s td_y (0.076)	rolling_mean _std_z (0.072)

BEST RF (rf)	s_y_mean (0.085)	s_x_mean (0.066)	s_z_mean (0.064)	s_y_energy (0.049)	s_y_std (0.031)
BEST RF (lgb)	s_y_mean (5138)	s_x_mean (3772)	s_z_mean (3204)	corr_xz (3029)	m_y_min (2973)
LGB + Residual CNN + augmented CNN ensemble (lgb)	s_y_mean (8984)	s_x_mean (6258)	s_z_mean (5466)	corr_xz (5182)	m_y_min (4943)
LightGBM 5- fold + single CNN (lgb)	s_y_mean (8984)	s_x_mean (6258)	s_z_mean (5466)	corr_xz (5182)	m_y_min (4943)
LGB + XGB + multi- seed CNN extended features (lgb) - first try	s_y_median (7398)	s_x_median (4648)	m_x_autocor r_lag30 (4549)	s_z_median (4449)	sma (4327)
LGB + XGB + multi- seed CNN extended	s_y_median (7398)	s_x_median (4648)	m_x_autocor r_lag30 (4549)	s_z_median (4449)	sma (4327)

features (lgb) - rerun					
RF + LGB 5- fold + multi- seed CNN (rf)	s_y_mean (0.085)	s_x_mean (0.066)	s_z_mean (0.064)	s_y_energy (0.049)	s_y_std (0.031)
RF + LGB 5- fold + multi- seed CNN (lgb)	s_y_mean (5138)	s_x_mean (3772)	s_z_mean (3204)	corr_xz (3029)	m_y_min (2973)
RF + LGB + Residua l CNN weight boosted (rf)	s_y_mean (0.085)	s_x_mean (0.066)	s_z_mean (0.064)	s_y_energy (0.049)	s_y_std (0.031)
RF + LGB + Residua l CNN weight boosted (lgb)	s_y_mean (5138)	s_x_mean (3772)	s_z_mean (3204)	corr_xz (3029)	m_y_min (2973)
CNN seed search (rf)	s_y_mean (0.085)	s_x_mean (0.066)	s_z_mean (0.064)	s_y_energy (0.049)	s_y_std (0.031)
CNN seed	s_y_mean (5138)	s_x_mean (3772)	s_z_mean (3204)	corr_xz (3029)	m_y_min (2973)

search (lgb)					
Person- independ ent LightG BM normali zed features	mag_norm_ zcr (2588)	mag_mean (2394)	corr_xz (2149)	ratio_y_std_ov er_range (2001)	mag_norm_i qr (1986)
Bigger CNN + label smoothi ng + spatial dropout (rf)	s_y_mean (0.085)	s_x_mean (0.066)	s_z_mean (0.064)	s_y_energy (0.049)	s_y_std (0.031)
Bigger CNN + label smoothi ng + spatial dropout (lgb)	s_y_mean (5138)	s_x_mean (3772)	s_z_mean (3204)	corr_xz (3029)	m_y_min (2973)

Q3 — CNN Training Behaviour

Method	Run	Final train loss	Final val loss	Train/v al gap	Final train acc	Final val acc	Epochs to conver ge	Overfit ?
Baseline RF	—	—	—	—	—	—	—	—

Hybrid RF + CNN-LSTM	R1	0.3394	— (no val)	—	0.8789	—	20/20	unknown
Hybrid RF + CNN-LSTM	R2	0.3432	— (no val)	—	0.8825	—	20/20	unknown
FFT + CNN-LSTM	R1	0.3374	— (no val)	—	0.8835	—	20/20	unknown
FFT + CNN-LSTM	R2	0.3414	— (no val)	—	0.8820	—	20/20	unknown
SMA + peak-frequency + CNN-LSTM	R1	0.3495	— (no val)	—	0.8799	—	25/25	unknown
SMA + peak-frequency + CNN-LSTM	R2	0.3529	— (no val)	—	0.8776	—	25/25	unknown
BEST RF + LGB + Residual CNN	R1	0.2256	0.3337	0.1081	0.9187	0.9038	34 (early stop, best ep.22)	Mild
BEST RF + LGB + Residual CNN	R2	0.1425	0.4037	0.2612	0.9465	0.8848	31 (early stop, best ep.19)	Yes
LGB + Residual CNN + augmented	R1	0.0057	0.0900	0.0843	0.9980	0.9746	60/60 (best ep.57)	Yes

d CNN (run 1)								
LGB + Residual CNN + augmente d CNN (run 2)	R1	0.2069	0.244 1	0.0372	0.926 0	0.926 5	12 (early stop, best ep.1)	Minim al
LGB + Residual CNN + augmente d CNN (run 1)	R2	0.0060	0.124 8	0.1188	0.997 9	0.971 0	53 (early stop, best ep.41)	Yes
LGB + Residual CNN + augmente d CNN (run 2)	R2	0.2101	0.245 4	0.0353	0.925 1	0.925 1	12 (early stop, best ep.1)	Mild
LightGB M 5-fold + single CNN	R1	0.1220	0.424 7	0.3027	0.954 1	0.892 9	34 (early stop, best ep.22)	Yes
LightGB M 5-fold + single CNN	R2	0.1195	0.492 0	0.3725	0.958 9	0.875 7	26 (early stop, best ep.14)	Yes
RF + LGB + Residual CNN weight boosted	R1	0.1474	0.492 8	0.3454	0.943 7	0.895 6	26 (early stop, best ep.14)	Yes
RF + LGB +	R2	0.1443	0.509 2	0.3649	0.945 9	0.886 6	24 (early	Yes

Residual CNN weight boosted							stop, best ep.12)	
CNN seed search	R1	best seed=2: val_loss=0.35 51, val_acc=0.89 66						Mild
CNN seed search	R2	best seed=0: val_loss=0.34 25, val_acc=0.89 93					11 epochs shown	Mild
Person- independ ent LightGB M normalize d features	R1/R 2	0.2613 (ep.11)	0.437 3	0.1760	0.909 3	0.889 3	11 (partial)	Mild
Bigger CNN + label smoothing + spatial dropout	R1	best seed=42: val_loss=0.35 16, val_acc=0.90 65						Mild
Bigger CNN + label smoothing + spatial dropout	R2	best seed=0: val_loss=0.28 98, val_acc=0.91 38						Mild

Q3 — LGB OOF CV F1 (5-fold, where available)

Method	Run	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	OOF mean
LGB + Residual CNN + augmented CNN ensemble	R1	0.7399	0.7100	0.7505	0.7410	0.7472	0.7380
LGB + Residual CNN + augmented CNN ensemble	R2	0.7300	0.7275	0.7101	0.7469	0.7307	0.7290
LightGBM 5-fold + single CNN	R1	0.7399	0.7100	0.7505	0.7410	0.7472	0.7380
LightGBM 5-fold + single CNN	R2	0.7300	0.7275	0.7101	0.7469	0.7307	0.7290
LGB + XGB + multi-seed CNN extended features (LGB)	R1	0.7548	0.7228	0.7551	0.7505	0.7737	0.7521
LGB + XGB + multi-seed CNN extended features (LGB)	R2	0.7537	0.7488	0.7171	0.7623	0.7636	0.7487
LGB + XGB + multi-seed CNN extended features (XGB)	R1	0.7425	0.7125	0.7388	0.7454	0.7574	0.7399
LGB + XGB + multi-seed CNN extended features (XGB)	R2	0.7559	0.7649	0.7108	0.7502	0.7402	0.7446
RF + LGB 5-fold + multi-seed CNN	R1	0.7336	0.7122	0.7416	0.7275	0.7432	0.7322
RF + LGB 5-fold + multi-seed CNN	R2	0.7335	0.7306	0.6984	0.7513	0.7366	0.7299
Person-independent LightGBM normalized features	R1	0.7380	0.7147	0.7387	0.7387	0.7492	0.7366
Person-independent LightGBM normalized features	R2	0.7515	0.7238	0.7020	0.7342	0.7443	0.7313

Method	Run	Macro -F1	L0 F1	L1 F1	L2 F1	L3 F1	L4 F1	L5 F1	Worst label
Baseline RF	R2	0.692 9	0.946 8	0.867 4	0.126 6	0.687 5	0.909 1	0.619 9	L2 (0.13)
BEST RF (rf)	R1/R 2	0.692 2	0.944 7	0.866 2	0.214 3	0.695 7	0.909 1	0.523 5	L2 (0.21)
BEST RF (lgb)	R1/R 2	0.712 0	0.958 3	0.890 8	0.179 8	0.694 7	0.909 1	0.639 5	L2 (0.18)
LGB + Residual CNN + augmented CNN (lgb)	R1/R 2	0.720 2	0.959 4	0.892 7	0.186 0	0.714 8	0.909 1	0.659 1	L2 (0.19)
LightGBM 5-fold + single CNN (lgb)	R1/R 2	0.720 2	0.959 4	0.892 7	0.186 0	0.714 8	0.909 1	0.659 1	L2 (0.19)
LGB + XGB + multi-seed CNN extended features (lgb)	R1	0.732 9	0.972 7	0.903 5	0.215 1	0.701 5	0.912 3	0.692 3	L2 (0.22)
LGB + XGB + multi-seed CNN extended features (lgb)	R2	0.732 9	0.972 7	0.903 5	0.215 1	0.701 5	0.912 3	0.692 3	L2 (0.22)
RF + LGB 5-fold +	R1/R 2	0.692 2	0.944 7	0.866 2	0.214 3	0.695 7	0.909 1	0.523 5	L2 (0.21)

multi-seed CNN (rf)									
RF + LGB 5-fold + multi-seed CNN (lgb_fold)	R1/R 2	0.712 0	0.958 3	0.890 8	0.179 8	0.694 7	0.909 1	0.639 5	L2 (0.18)
RF + LGB + Residual CNN weight boosted (rf)	R1/R 2	0.692 2	0.944 7	0.866 2	0.214 3	0.695 7	0.909 1	0.523 5	L2 (0.21)
RF + LGB + Residual CNN weight boosted (lgb)	R1/R 2	0.712 0	0.958 3	0.890 8	0.179 8	0.694 7	0.909 1	0.639 5	L2 (0.18)
CNN seed search (rf)	R1/R 2	0.692 2	0.944 7	0.866 2	0.214 3	0.695 7	0.909 1	0.523 5	L2 (0.21)
CNN seed search (lgb)	R1/R 2	0.712 0	0.958 3	0.890 8	0.179 8	0.694 7	0.909 1	0.639 5	L2 (0.18)
Person- independe nt LightGBM normalized features	R1	0.720 6	0.960 5	0.895 8	0.226 9	0.698 5	0.881 7	0.660 0	L2 (0.23)
Person- independe nt LightGBM normalized features	R2	0.707 6	0.967 2	0.899 9	0.172 0	0.691 4	0.892 9	0.622 2	L2 (0.17)

Bigger CNN + label smoothing + spatial dropout (rf)	R1/R 2	0.692 2	0.944 7	0.866 2	0.214 3	0.695 7	0.909 1	0.523 5	L2 (0.21)
Bigger CNN + label smoothing + spatial dropout (lgb)	R1/R 2	0.712 0	0.958 3	0.890 8	0.179 8	0.694 7	0.909 1	0.639 5	L2 (0.18)

Q4 — Dominant Confusion Pairs

Method	Run	Confusion pair 1	Confusion pair 2
Baseline RF	R2	L2→L1 (58/72)	L5→L1 (42/105)
Hybrid RF + CNN-LSTM	R1/R2	— (rf not found)	
FFT + CNN-LSTM	R1/R2	— (rf not found)	
SMA + peak-freq + CNN-LSTM	R1/R2	— (rf not found)	
BEST RF (rf)	R1/R2	L2→L1 (50/72)	L5→L1 (55/105)
BEST RF (lgb)	R1/R2	L2→L1 (48/72)	L5→L1 (39/105)
LGB + Residual CNN + augmented CNN (lgb)	R1/R2	L2→L1 (49/72)	L5→L1 (38/105)
LightGBM 5-fold + single CNN (lgb)	R1/R2	L2→L1 (49/72)	L5→L1 (38/105)
LGB + XGB + multi-seed CNN extended features (lgb)	R1/R2	L2→L1 (47/72)	L5→L1 (30/105)
RF + LGB 5-fold + multi-seed CNN (rf)	R1/R2	L2→L1 (50/72)	L5→L1 (55/105)
RF + LGB 5-fold + multi-seed CNN (lgb_fold)	R1/R2	L2→L1 (48/72)	L5→L1 (39/105)
RF + LGB + Residual CNN weight boosted (rf)	R1/R2	L2→L1 (50/72)	L5→L1 (55/105)
RF + LGB + Residual CNN weight boosted (lgb)	R1/R2	L2→L1 (48/72)	L5→L1 (39/105)

CNN seed search (rf)	R1/R2	L2→L1 (50/72)	L5→L1 (55/105)
CNN seed search (lgb)	R1/R2	L2→L1 (48/72)	L5→L1 (39/105)
Person-independent LightGBM normalized features	R1	L2→L1 (48/358 OOF)	L5→L3 (34/526 OOF)
Person-independent LightGBM normalized features	R2	L2→L1 (48/72)	L5→L1 (34/105)
Bigger CNN + label smoothing + spatial dropout (rf)	R1/R2	L2→L1 (50/72)	L5→L1 (55/105)
Bigger CNN + label smoothing + spatial dropout (lgb)	R1/R2	L2→L1 (48/72)	L5→L1 (39/105)